

What is quantization?

Quantization is a model optimization technique that reduces the precision of numerical values such as weights and activations in models to make them faster and more efficient. It helps lower memory usage, model size, and computational cost while maintaining almost the same level of accuracy.

Why quantization?

- **Reduced Model Size:** Using smaller data types requires fewer bits per parameter, significantly lowering memory usage.
- **Faster Inference:** Lower-precision computations (like INT8 or INT4) execute faster on CPUs, GPUs and specialized hardware.
- **Edge Deployment:** Quantized models can efficiently run on low-power devices such as mobiles and embedded systems.
- **Lower Operational Costs:** By cutting down on power and memory needs, quantization helps minimize overall infrastructure expenses.

How It Works?

Convert high-precision numbers like floats into lower-precision integers (like INT8).

The process revolves around two key parts:

- **Weight Quantization:** Compresses the trained model's parameters.
- **Activation Quantization:** Compresses the outputs generated during inference.

Methods of quantization:

- **Post-Training Quantization (PTQ):**
Quantization is applied after the model is fully trained, converting high-precision parameters (like FP32 weights and activations) into lower-bit formats such as INT8 to reduce computational load and memory usage.
Achieved using two primary techniques: **Static Quantization** which is pre-calibrated with representative data and **Dynamic Quantization** where activations are quantized on the fly during inference.

Pros: Fast, no retraining. **Cons:** Higher accuracy loss.

- **Quantization-Aware Training (QAT):**
The model is trained with quantization in mind, simulating low-precision operations during training. Helps the model adapt to quantization effects and minimize accuracy loss.

Pros: High accuracy, recovers lost precision. **Cons:** Computationally expensive.